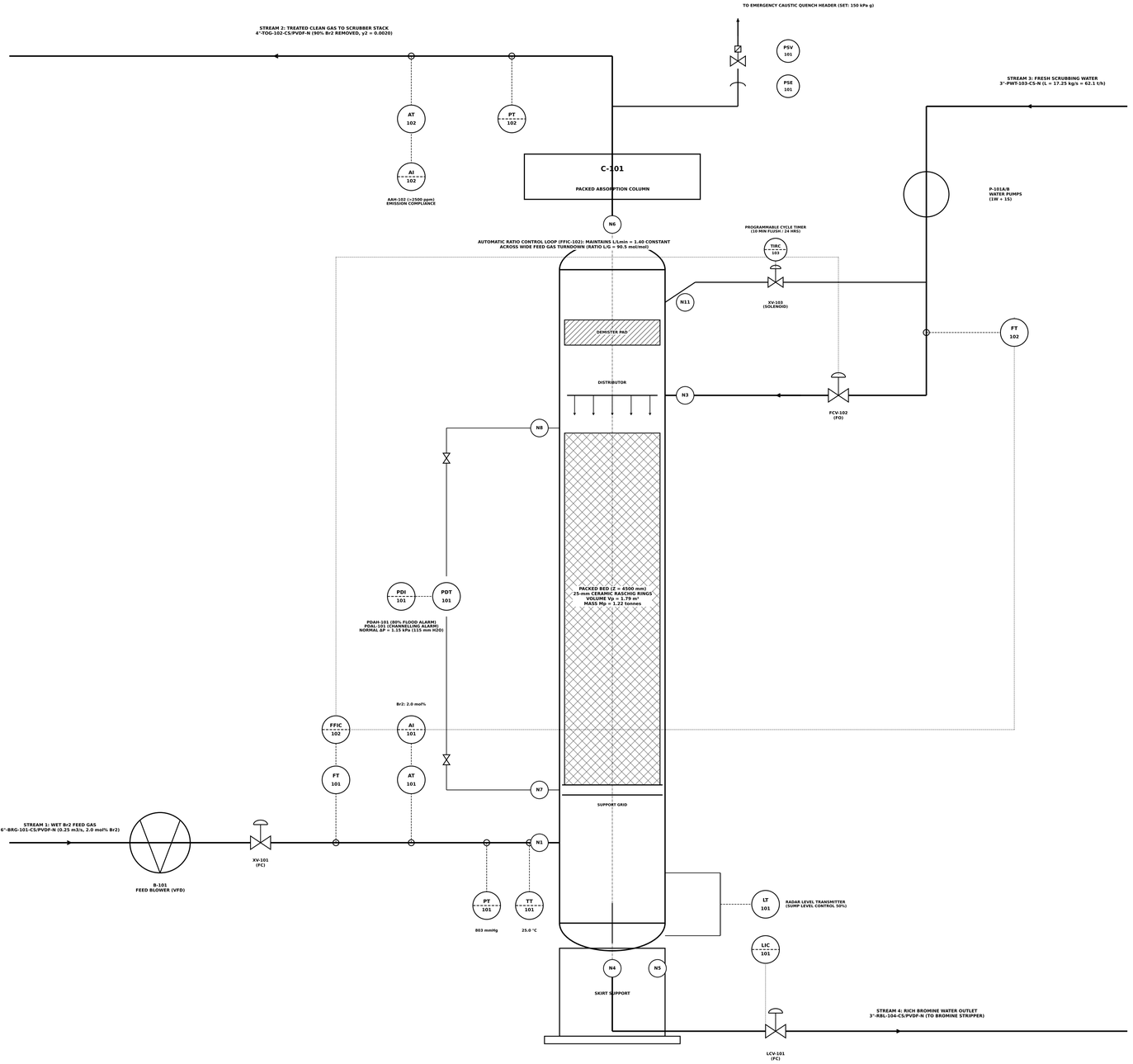


THIRD ANGLE PROJECTION

TABLE 1 — STREAM OPERATING MASS & HEAT BALANCE TABLE

STR	STREAM SERVICE	FLUID / PHASE	MASS FLOW	TEMP	PRESS	COMPOSITION
1	Feed Gas Mixture	Gas / Vapor	0.341 kg/s (1228 kg/h)	25 °C	803 mmHg	2.00 mol% Br2, 98% Air
2	Treated Overhead Gas	Gas / Vapor	0.315 kg/s (1134 kg/h)	25 °C	794 mmHg	0.20 mol% Br2 (90% Rem)
3	Scrubbing Water Feed	Liquid / H2O	17.25 kg/s (62.1 t/h)	25 °C	2.5 bar g	100.0 wt% Pure Water
4	Rich Liquor Bottoms	Liquid Soln	17.28 kg/s (62.2 t/h)	25 °C	1.1 bar g	x1 = 2.79x10⁻² (Br2)
5	Safety Relief Vent	Emergency	Discontinuous / Upset	25 °C	1.5 bar g	Raw Gas / Relief Quench
6	Demister Wash Water	Liquid / H2O	2.50 kg/s (Batch Flush)	25 °C	3.0 bar g	Intermittent Washdown

TABLE 2 — PIPING LINE DESIGNATION & SPECIFICATION SCHEDULE

LINE NUMBER	SERVICE DESCRIPTION	SIZE	CLASS	PIPE MATERIAL	INSULATION
6"-BRG-101-CS/PVDF-N	Raw Bromine Gas Inlet	**6" (DN150)**	CL 150	SA-106 Gr B + 3 mm PVDF	None
4"-TOG-102-CS/PVDF-N	Treated Gas Vent to Stack	**4" (DN100)**	CL 150	SA-106 Gr B + 3 mm PVDF	None
3"-PWT-103-CS-N	Fresh Water Supply Header	3" (DN80)	CL 150	Carbon Steel SA-106 Gr B	None
3"-RBL-104-CS/PVDF-N	Rich Bromine Water Bottoms	3" (DN80)	CL 150	SA-106 Gr B + 3 mm PVDF	None
1.5"-DRN-105-CS/PVDF-N	Vessel Drain Line	**1.5" (DN40)**	CL 150	SA-106 Gr B + 3 mm PVDF	None
1.5"-DMW-106-SS316L-N	Demister Flush Water	**1.5" (DN40)**	CL 150	SS316L Stainless Steel	None

TABLE 3 — INSTRUMENTATION & AUTOMATIC CONTROL VALVES SCHEDULE

TAG	DESCRIPTION	TYPE / TECHNOLOGY	RANGE / SETPOINT	FAIL ACTION
FFIC-102	Water-to-Gas Ratio Controller	DCS Ratio Algorithm	L/G = 90.5 mol/mol	Auto Tracking
FCV-102	Water Supply Control Valve	Globe / Diaphragm Act.	0 - 25 kg/s Water	Fail-Open (FO)
LIC-101	Bottom Sump Level Controller	DCS PID Loop	Set: 50% Sump Height	Auto Control
LCV-101	Rich Liquor Discharge Valve	Globe / PTFE Lined	0 - 25 kg/s Liquor	Fail-Closed (FC)
PDT-101	Packed Bed Diff. Pressure	Diaphragm Seal (Hast)	0 - 5.0 kPa (0-500 mm)	Alarm High/Low
XV-101	Feed Gas Isolation Valve	Pneumatic Ball Valve	Tight Shutoff (TSO)	Fail-Closed (FC)
PSV-101	Emergency Safety Relief Valve	Conventional Flanged	Set: 150 kPa (g)	Relief to Quench

PROCESS CONTROL PHILOSOPHY & SAFETY INTERLOCK SYSTEM (SIS)

1. GAS-LIQUID RATIO CONTROL STRATEGY (FFIC-102):

- FLOW OF SCRUBBING WATER (FCV-102) IS CONTINUOUSLY SLAVED TO INLET GAS MASS FLOW (FT-101).
- **RATIO MAINTAINED AT L = 1.40 Lmin (L/G = 90.5 mol/mol). IF GAS FLOW INCREASES, WATER FLOW**
- RAMPS UP PROPORTIONATELY TO GUARANTEE 90% BROMINE CAPTURE EFFICIENCY AT ALL TIMES.

2. BED DIFFERENTIAL PRESSURE FLOODING PROTECTION (PDT-101):

- **NORMAL OPERATING BED PRESSURE DROP: 1.15 kPa (115 mm H2O).**
- **HIGH ALARM (PDAH-101 AT 2.5 kPa): WARNS OPERATOR OF INCIPIENT BED FLOODING.**
- **HIGH-HIGH INTERLOCK (PDAHH-101 AT 3.8 kPa): AUTOMATICALLY TRIPS FEED BLOWER B-101 AND**
- CLOSES XV-101 TO PREVENT BROMINE LIQUID CARRYOVER INTO OVERHEAD DUCTWORK.

3. LEVEL & OVERFLOW INTERLOCKS (LT-101 / LIC-101):

- LOW-LOW LEVEL (LSLL-101): TRIPS BOTTOM PUMPS TO PREVENT CAVITATION & GAS BLOW-THROUGH.
- HIGH-HIGH LEVEL (LSHH-101): SOUNDS CRITICAL EVAPORATOR/ABSORBER ALARM IN MAIN DCS CONTROL ROOM.

JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS EQUIPMENT DESIGN & DRAWING

DRAWING TITLE PACKED-BED ABSORBER SYSTEM (C-101) — PIPING & INSTRUMENTATION DIAGRAM (P&ID) CAD	**LAB / SUBJECT** PROCESS EQUIPMENT DESIGN & DRAWING	**COURSE / BRANCH** B.E. CHEMICAL ENGINEERING	**SCALE & REVISION** SCALE: DIAGRAMMATIC / ISA-5.1 STANDARD | REV: 00
STUDENT NAME SOURISH SENAPATI	**EXAM ROLL NUMBER** 002310301035	**DRAWING / DOCUMENT NO.** PEDD-PBA-05 (SHEET 05 OF 05)	**DATE** 29 / 09 / 2026
EQUIPMENT IDENTIFICATION & SERVICE PACKED BED ABSORBER PROCESS PIPING & AUTOMATIC INSTRUMENTATION	**DESIGN BASIS & METALLURGY / LINING** ASME SEC VIII DIV 1 / SA-516 GR 70 + PVDF LINER / SS316L / CERAMIC	**PROCESS PERFORMANCE & RECOVERY** FEED: 0.25 m³/s (2% Br2) | WATER: 17.25 kg/s | 90% RECOVERY | RATIO L/G = 90.5	**DESIGN STATUS CLASSIFICATION** CALCULATED: ID 750, Z 4500 | PRELIMINARY CAD: SHELL & NOZZLES

DIMENSIONS: METRIC (MM / M) | FLANGES: ASME B16.5 CLASS 150 RF | WELDING: ASME SEC IX | THIRD ANGLE PROJECTION